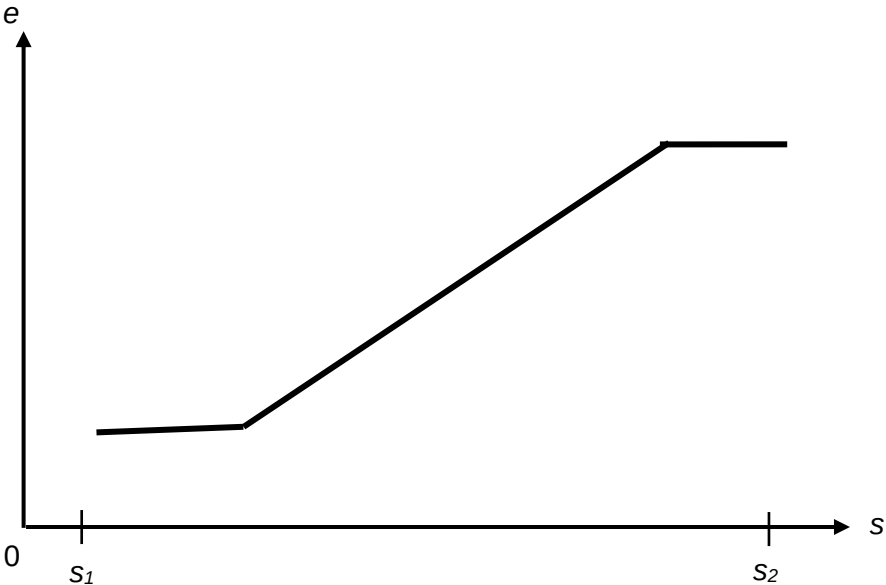


	$\rho = 1.745 \text{ g} = 1.7 \text{ g}$	
2a	The object is in equilibrium and hence resultant force is zero. Before lowering into water, $F_s = mg$ After lowering into water $F_s' + Up = mg$ $F_s' + Up = F_s$ $Up = F_s - F_s' = k(x' - x)$ Hence, upthrust = decrease in spring force $= k\Delta x$ [Note: cannot just be substituting numbers, but must indicate it is representing Δx] $= 42 (0.0045)$ $= 0.189 \text{ N}$ $= 0.19 \text{ N (to 2 s.f.)}$	M1 A1
2b	Upthrust = $V\rho g$ $V(1000)(9.81) = 0.19$ $V = 1.94 \times 10^{-5} \text{ m}^3$	M1 A1

2c	By Newton's 3rd Law, force of water on mass = - force of mass on water. Since force of water on mass is the upthrust, magnitude of force of mass on water is therefore same magnitude as upthrust. Hence magnitude of force of mass on water = 0.19 N	M1 A1
2d	When fully inside water, $F_s + V\rho g = mg$ $ke + V\rho g = mg$ $e = \frac{1}{k}(mg - V\rho g)$ So, first part of graph while inside water is a horizontal flat line since e is independent of s. While it is moving out of water, $F_s + V\rho g = mg$ $ke + V\rho g = mg$ $e = \frac{1}{k}(mg - LA\rho g)$ L is the submerged height of the mass and thus varies linearly with s. So, the second part of the graph is a linear line with increasing e. When the mass is fully out of water, the spring force is thus equal to weight and there is no upthrust, so the graph is a horizontal flat line independent of s. 	B1 B1 B1
2e	The gravitational potential at a point is the work done per unit mass	B1